ADVANCING GENERALIZABILITY IN BRAIN TUMOR SEGMENTATION ACROSS DIVERSE POPULATIONS

Team Members:

Ahmed Samy Mohsen (Computer and Systems Engineering)

Omar Mohammed Eldash (Biomedical Engineering)

Mohamed Hany Gabr (Biomedical Engineering)

Under Supervision of:

Dr Rasha Ali (Computer and Systems Engineering)

Dr Rana Hossam(Biomedical Engineering)

06/07/2026

Introduction

- Brain tumor segmentation is essential for identifying tumor boundaries in MRI scans, supporting diagnosis, treatment planning, and disease monitoring.

- Manual annotation by expert radiologists is very accurate but slow, laborintensive, and costly, especially in low resource settings.

- Deep learning offers a faster, more consistent alternative, But most models lack generalizability across domains.

Clinical Urgency

5th

Most Common Cancer

Brain tumors rank among the top cancer types globally

1M+

Patients in the U.S.

Individuals currently living with a brain tumor

12%

5-Year Survival Rate

Adults surviving five years post-diagnosis

10th

Leading Cause of Death

For both men and women worldwide

Manual segmentation by radiologists takes **3–5 hours per case** — a critical bottleneck that delays diagnosis and strains clinical resources.

Problem Statement

Domain Generalizability Gap

Scanner variability, protocol differences, and tumor heterogeneity cause models to fail across populations.

High Computational Cost

Transformer-based methods demand heavy GPU resources, blocking real-time clinical deployment.

Manual Labor Intensity

Expert annotation requires 3–5 hours per case — costly and unscalable, especially in low-resource settings.

Class Imbalance

Unequal population sizes introduce bias, degrading fairness and accuracy for underrepresented cohorts.

Gap Analysis

- **Dependency on Perfect Data**: Most solutions break down if one MRI modality is missing (common in clinical routine) or if the image resolution varies between the 5 cohorts. (T1, T1ce, T2, FLAIR)

- **High Computational Cost**: Transformer-based methods that improve generalization often require heavy GPU resources, making them unsuitable for real-time clinical deployment in resource-constrained hospitals.

Objectives

- Develop a single, generalizable, resource-efficient 3D brain tumor segmentation model.
- Apply Fairness strategy to solve unbalanced dataset problem.
- Develop an application to utilize the strength of our model and visualize 3D segmentation mask.

Dataset

- MRI data varies widely across scanners, hospitals, and patient groups, and tumors themselves differ in shape, size, and appearance. Our study focuses on five distinct tumor populations, each with unique challenges.

- The BraTS (Brain Tumor Segmentation) dataset is the industry standard for evaluating medical image segmentation models.

- Standard Volume: Typically provided as 240 x 240 x 155 voxels in NIfTI (.nii) format.

- Multimodal MRI: Every patient case includes 4 distinct ,skull-stripped scans:

    1. T1-weighted (T1): Best for anatomical structure.
    2. T1-Contrast enhanced (T1ce): Highlights the active, enhancing tumor core.
    3. T2-weighted (T2): Visualizes the core and edema.
    4. Fluid-Attenuated Inversion Recovery (FLAIR): Best for detecting peritumoral swelling (edema).

Dataset

Tumor Sub-Regions & Target Populations

Nested Segmentation Labels

ET — Enhancing Tumor

Active tumor core; bright on T1ce

TC — Tumor Core

ET + Necrotic Core (NC)

WT — Whole Tumor

ET + NC + Peritumoral Edema (ED)

Five Target Populations

GLI

Adult Glioma

SSA

Sub-Saharan Glioma

PED

Pediatric Glioma

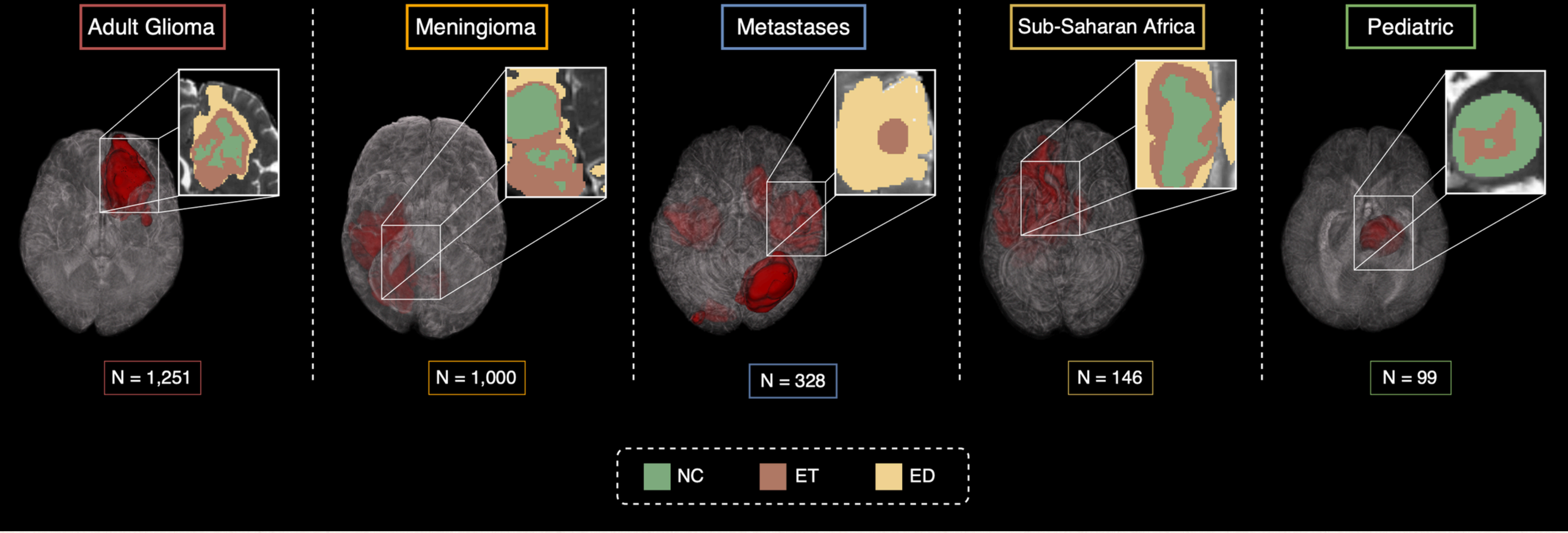
MEN

Meningioma

MET

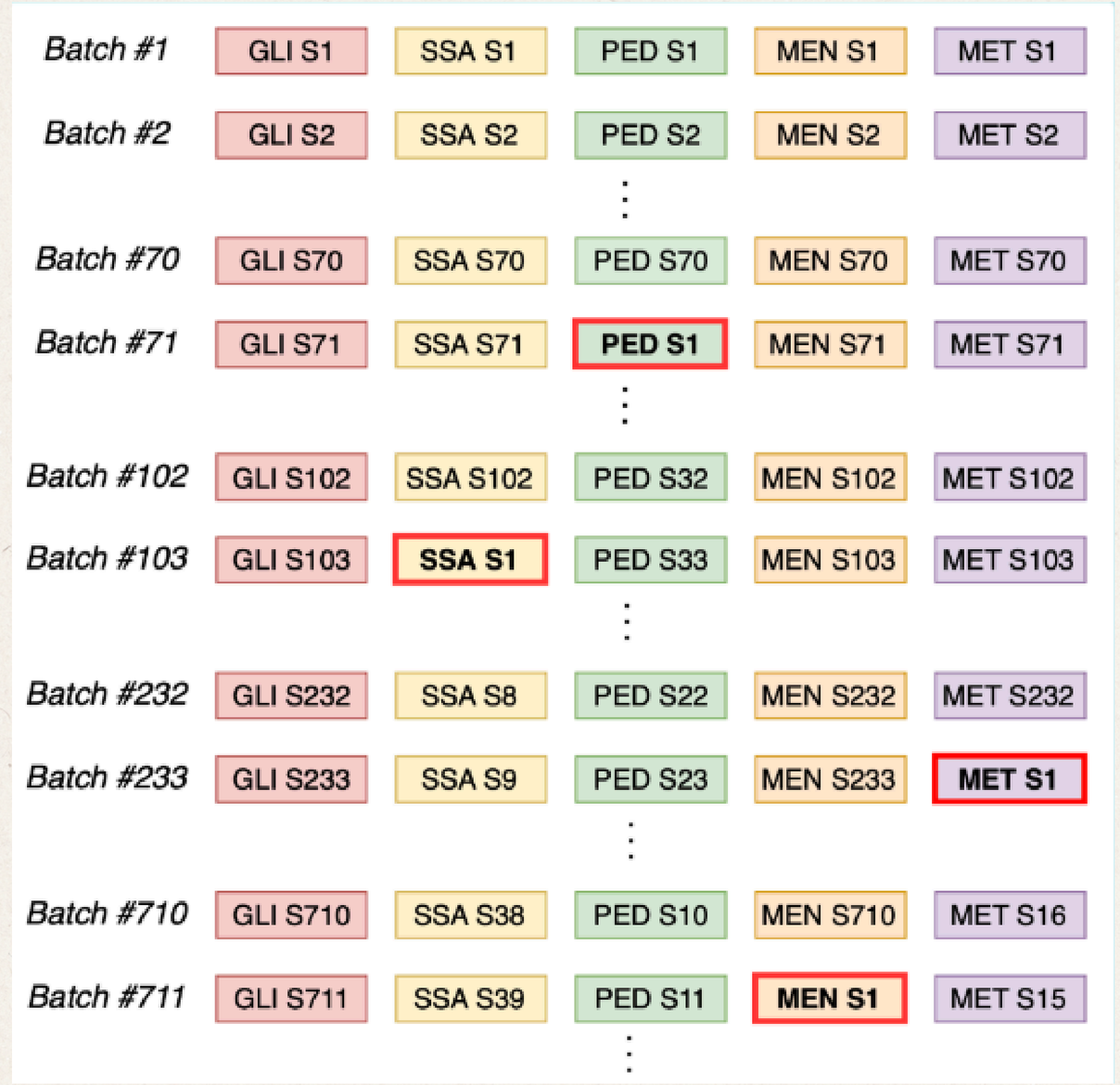
Metastases

These populations vary in scanner quality, tumor morphology, and anatomical distribution — the core generalization challenge.



- Representative examples showcasing the morphological and anatomical diversity of brain tumors across the five population groups used in this study.
- These examples illustrate significant variations in tumor size, shape, location, and spatial distribution, showing the inherent challenge of designing a single, generalizable segmentation model.
- The sample count for each population is denoted by N.

Stratified Batch Sampling



Mentorship and Sponsorship

Academic Mentor

Prof. Mohamed Al-Masni

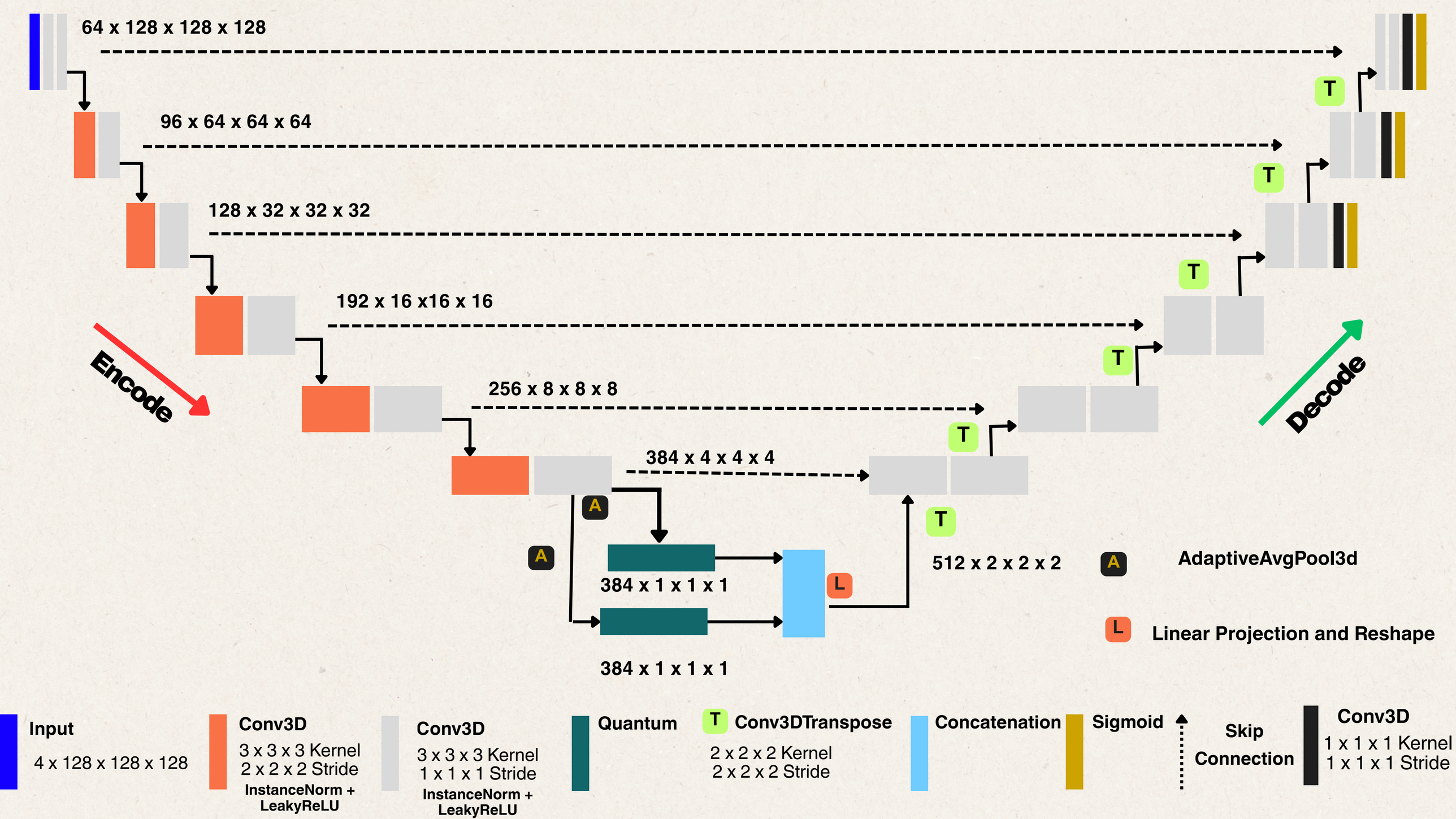
Sejong University, Seoul, South Korea

Sponsor / Collaboration

Colab Pro Plus

Architecture

- Core Model: DynUNet (Dynamic U-Net) as the backbone for 3D spatial feature extraction.
- Innovation: Parallel Quantum Bottleneck:
  - Replaces the deepest layer with a Quantum Layer using PennyLane.
  - Parallel execution paths (Quantum Layer 1 & 2).


64 x 128 x 128 x 128
96 x 64 x 64 x 64
128 x 32 x 32 x 32
192 x 16 x16 x 16
256 x 8 x 8 x 8
384 x 4 x 4 x 4
Encode
Decode
A
T
L
384 x 1 x 1 x 1
384 x 1 x 1 x 1
512 x 2 x 2 x 2
AdaptiveAvgPool3d
Linear Projection and Reshape
Input
4 x 128 x 128 x 128
Conv3D
3 x 3 x 3 Kernel
2 x 2 x 2 Stride
InstanceNorm + LeakyReLU
Conv3D
3 x 3 x 3 Kernel
1 x 1 x 1 Stride
InstanceNorm + LeakyReLU
Quantum
Conv3DTranspose
2 x 2 x 2 Kernel
2 x 2 x 2 Stride
Concatenation
Sigmoid
Skip Connection
Conv3D
1 x 1 x 1 Kernel
1 x 1 x 1 Stride

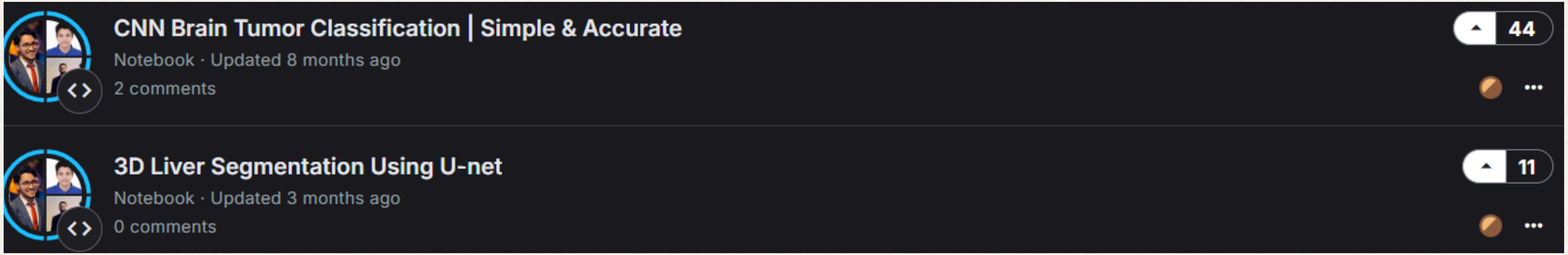
Proof of concept

Dataset: Decathlon dataset Task03

3D Liver Segmentation

	Classical Model	Hybrid Model	Difference (%)
Train Dice	0.9262	0.9332	**+0.76%**
Validation Dice	0.9229	0.9360	**+1.42%**
Total Parameters	137,662,753	66,922,971	**-51.39%**

Proof of concept



Knowledge Distillation

Goal

Train a lightweight student model that achieves high segmentation accuracy with lower computational cost.

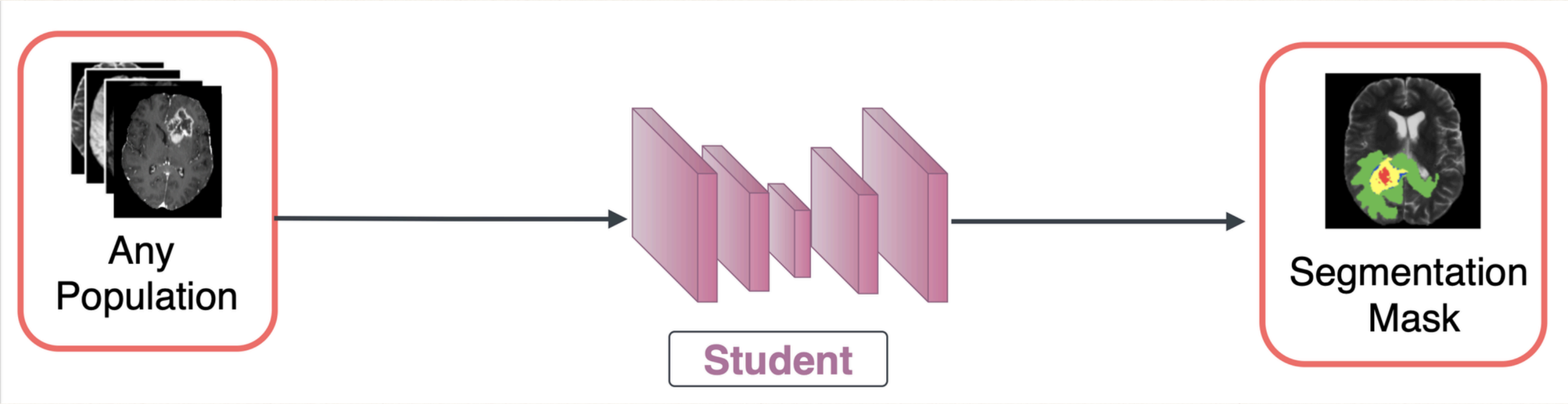
What are we doing?

Use a large Teacher model to guide a small Student model.
Both models receive the same input image.
The student learns from ground truth labels and from the teacher’s knowledge.

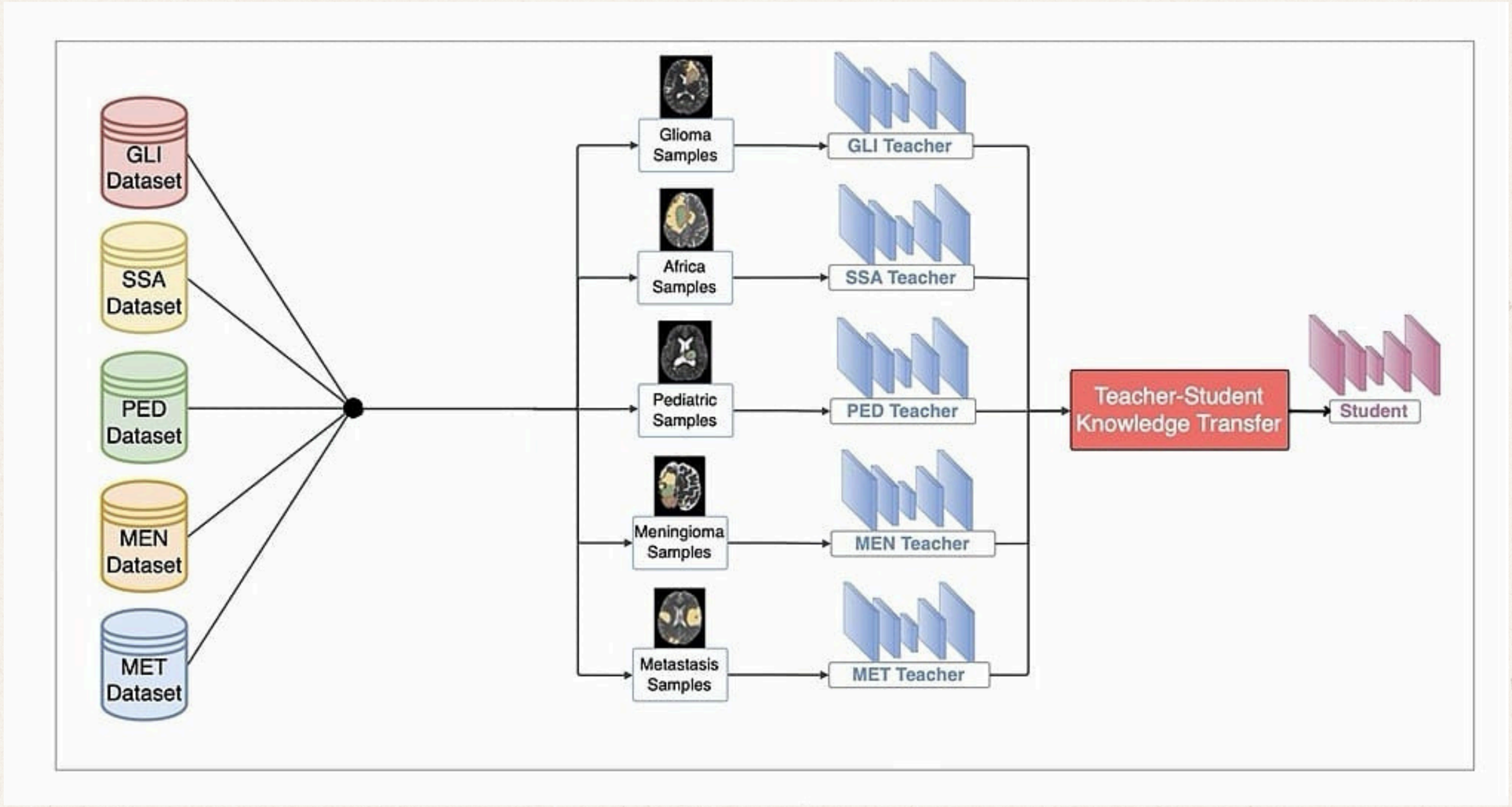
Why?

To obtain high accuracy with faster inference.
To make the model suitable for real-time and resource-limited applications.

Proposed Solution - Generalized Student Model



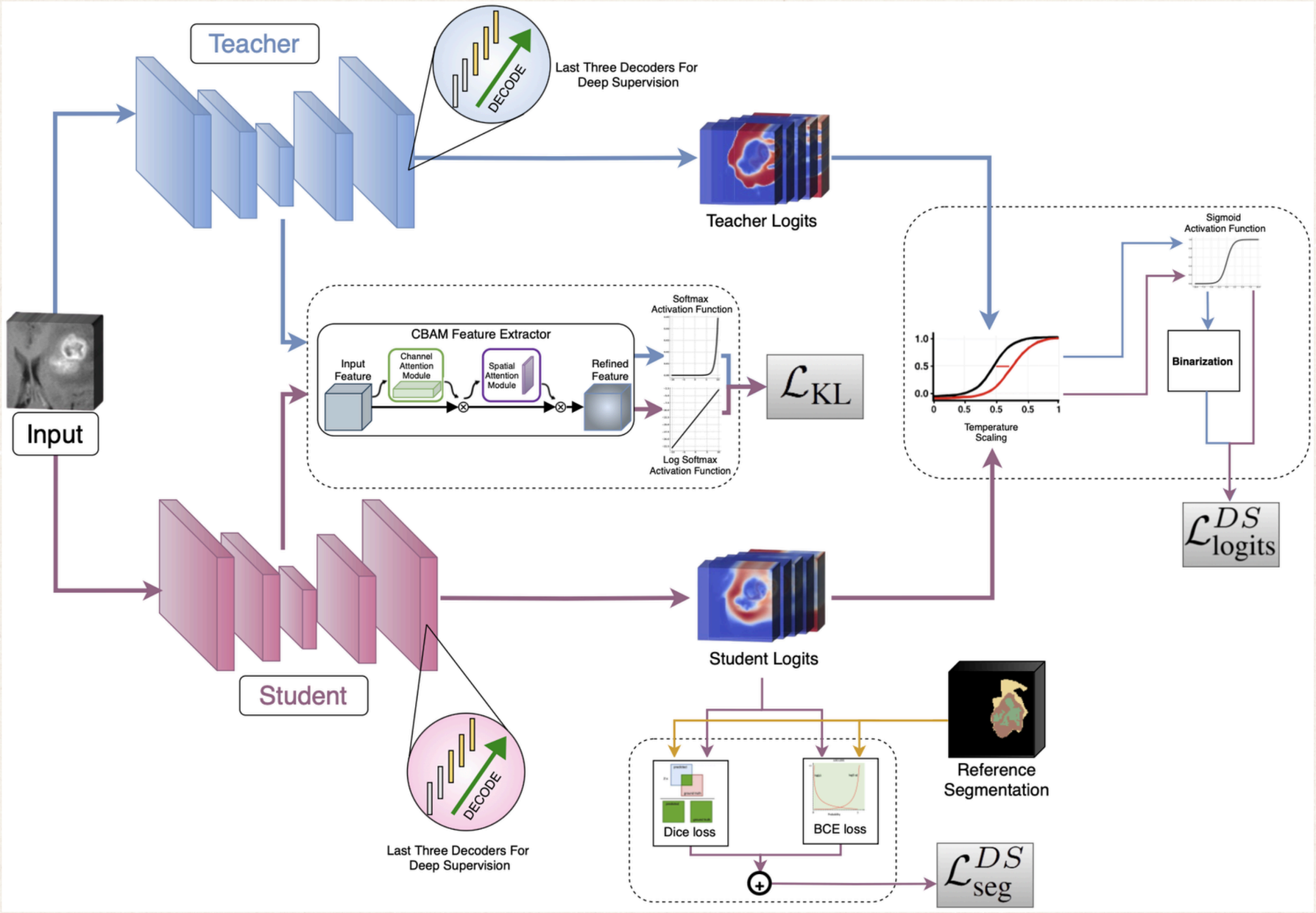
Knowledge Distillation



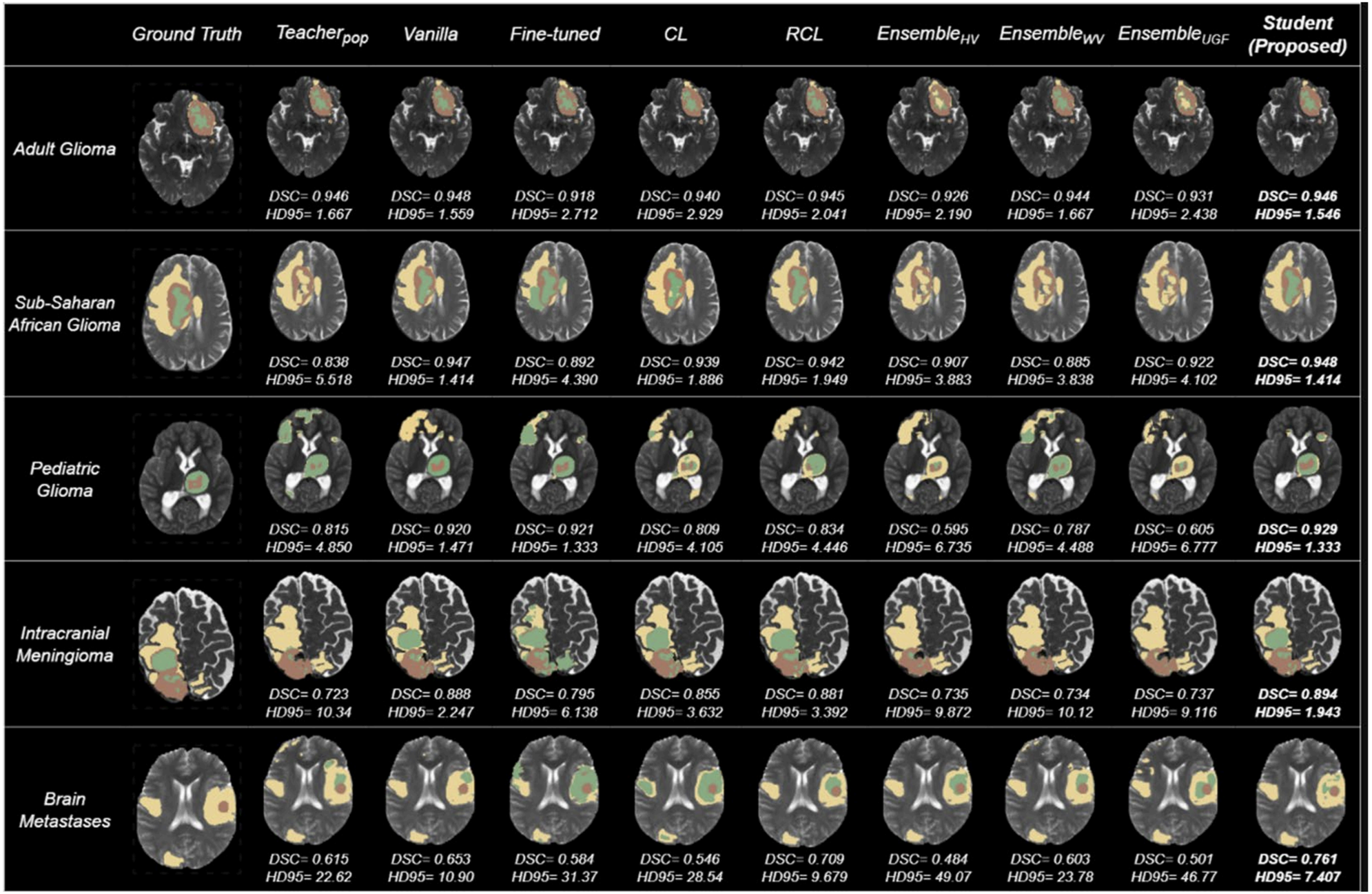
Teacher Models Comparison

	Metrics	Hybrid Teacher	Classical Teacher	Percentage Difference
Glioma	Average Dice	0.8864	0.8823	**0.46%**
	Average HD95	6.5897	6.2037	**6.22%**
SSA	Average Dice	0.7906	0.8013	**-1.34%**
	Average HD95	14.4235	13.5077	**6.78%**
PED	Average Dice	0.7162	0.6587	**8.73%**
	Average HD95	20.261	35.5233	**-42.96%**
MEN	Average Dice	0.8262	0.8247	**0.18%**
	Average HD95	24.888	35.3433	**-29.58%**
MET	Average Dice	0.6037	0.5627	**7.29%**
	Average HD95	27.5596	47.87	**-42.43%**

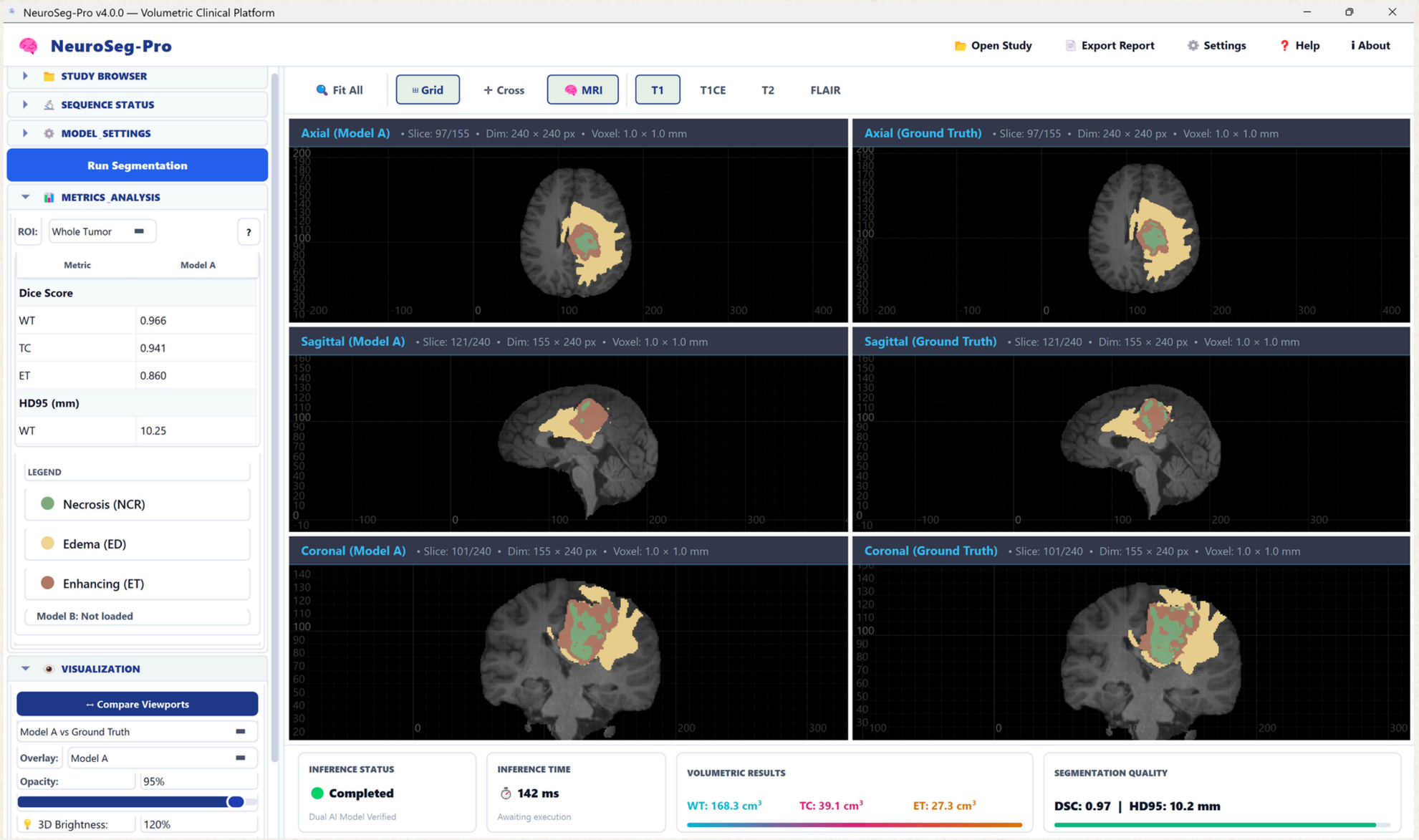
Knowledge Distillation



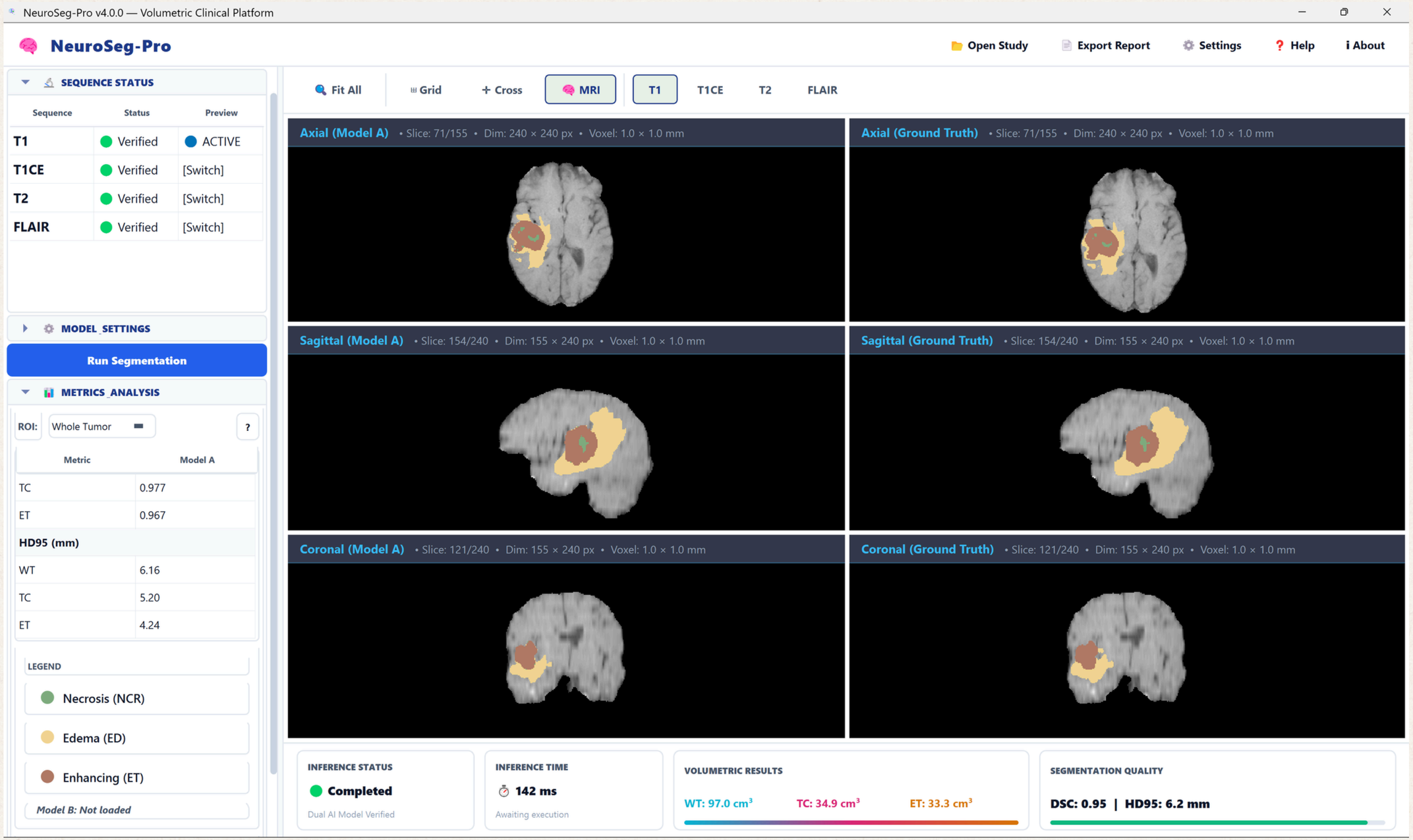
Comparison


Ground Truth
Teacher$_{pop}$
Vanilla
Fine-tuned
CL
RCL
Ensemble$_{HV}$
Ensemble$_{WV}$
Ensemble$_{UGF}$
Student (Proposed)
Adult Glioma
DSC= 0.946 HD95= 1.667
DSC= 0.948 HD95= 1.559
DSC= 0.918 HD95= 2.712
DSC= 0.940 HD95= 2.929
DSC= 0.945 HD95= 2.041
DSC= 0.926 HD95= 2.190
DSC= 0.944 HD95= 1.667
DSC= 0.931 HD95= 2.438
DSC= 0.946 HD95= 1.546
Sub-Saharan African Glioma
DSC= 0.838 HD95= 5.518
DSC= 0.947 HD95= 1.414
DSC= 0.892 HD95= 4.390
DSC= 0.939 HD95= 1.886
DSC= 0.942 HD95= 1.949
DSC= 0.907 HD95= 3.883
DSC= 0.885 HD95= 3.838
DSC= 0.922 HD95= 4.102
DSC= 0.948 HD95= 1.414
Pediatric Glioma
DSC= 0.815 HD95= 4.850
DSC= 0.920 HD95= 1.471
DSC= 0.921 HD95= 1.333
DSC= 0.809 HD95= 4.105
DSC= 0.834 HD95= 4.446
DSC= 0.595 HD95= 6.735
DSC= 0.787 HD95= 4.488
DSC= 0.605 HD95= 6.777
DSC= 0.929 HD95= 1.333
Intracranial Meningioma
DSC= 0.723 HD95= 10.34
DSC= 0.888 HD95= 2.247
DSC= 0.795 HD95= 6.138
DSC= 0.855 HD95= 3.632
DSC= 0.881 HD95= 3.392
DSC= 0.735 HD95= 9.872
DSC= 0.734 HD95= 10.12
DSC= 0.737 HD95= 9.116
DSC= 0.894 HD95= 1.943
Brain Metastases
DSC= 0.615 HD95= 22.62
DSC= 0.653 HD95= 10.90
DSC= 0.584 HD95= 31.37
DSC= 0.546 HD95= 28.54
DSC= 0.709 HD95= 9.679
DSC= 0.484 HD95= 49.07
DSC= 0.603 HD95= 23.78
DSC= 0.501 HD95= 46.77
DSC= 0.761 HD95= 7.407

Student Results



Student Results



References

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2015, pp. 234–241.

[2] O. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning dense volumetric segmentation from sparse annotation," in MICCAI, 2016, pp. 424–432.

[3] F. Milletari, N. Navab, and S. A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in IEEE 3D Vision (3DV), 2016, pp. 565–571.
s on Medical Imaging, vol. 34, no. 10, pp. 1993–2024, 2015.

[4] A. Myronenko, "3D MRI brain tumor segmentation using autoencoder regularization," in International Brain Tumor Segmentation (BraTS) Challenge, 2018, pp. 311–320.

[5] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," Nature Methods, vol. 18, no. 2, pp. 203–211, 2021.

[6] A. Hatamizadeh et al., "UNETR: Transformers for 3D medical image segmentation," in IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022, pp. 1748–1758.

[7] A. Hatamizadeh et al., "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in International MICCAI Brainlesion Workshop, 2022, pp. 272–284.

[8] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," arXiv preprint arXiv:1503.02531, 2015.

[9] M. J. Cardoso et al., "MONAI: An open-source framework for deep learning in healthcare," arXiv preprint arXiv:2211.02701, 2022.

[10] B. H. Menze et al., "The multimodal brain tumor image segmentation benchmark (BraTS)," IEEE Transaction

Thank you

Questions Time